E-Commerce Customer Churn Analysis

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Executive Summary & Problem Statement

In the highly competitive e-commerce landscape, understanding customer churn patterns is critical for ensuring customer satisfaction, driving long-term retention, and sustaining profitability. Customer acquisition costs significantly outweigh retention expenses; thus, identifying key churn drivers enables proactive intervention.

This project utilizes historical transactional and behavioral data from the `ecomm` database to investigate customer attrition dynamics. By examining customer attributes—including tenure, preferred payment modes, satisfaction scores, complaint history, order patterns, and return behaviors—this analysis uncovers underlying churn drivers. The ultimate objective is to equip e-commerce leadership with actionable, data-backed insights to deploy targeted retention strategies and enhance customer lifetime value.

Technical Architecture & File Structure

The project implementation is organized into two sequential SQL execution scripts:

> **`01_ecomm_data.sql`**
> Database instantiation, schema creation, raw dataset insertion, missing value imputation (mean/mode), outlier mitigation, categorical standardization, and schema transformation (column renaming, feature derivation, and dropping redundant flags).

> **`02_ecomm_data_exploration.sql`**
> Comprehensive analytical exploration addressing specific strategic business questions, creation of the customer_returns table, and relational joins analyzing customer returns alongside complaint and churn data.

Project Objectives & Implementation Workflow

1. Data Cleaning

- **Missing Value Imputation:** Calculated rounded continuous means to impute missing entries in WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, and DaySinceLastOrder. Applied discrete statistical modes to populate missing entries in Tenure, CouponUsed, and OrderCount.
- **Outlier Handling:** Identified and removed records exhibiting extreme anomalous distances where WarehouseToHome exceeded 100 km.

- **Categorical Standardization:** Consolidated fragmented device names by replacing occurrences of "Phone" in PreferredLoginDevice and "Mobile" in PreferedOrderCat with "Mobile Phone". Standardized payment method nomenclature by mapping "COD" to "Cash on Delivery" and "CC" to "Credit Card" in PreferredPaymentMode.

2. Data Transformation

- **Column Renaming:** Renamed PreferedOrderCat to PreferredOrderCat for proper spelling and consistency, and renamed HourSpendOnApp to HoursSpentOnApp for clarity.
- **Derived Columns & Feature Engineering:** Created ComplaintReceived ("Yes" / "No") mapped from the binary Complain flag, and created ChurnStatus ("Churned" / "Active") mapped from the binary Churn flag.
- **Schema Optimization:** Dropped the legacy binary columns Churn and Complain to prevent redundancy and establish clean reporting standards.

3. Data Exploration & Analytical Findings

The second script (02_ecomm_data_exploration.sql) answers core operational and analytical questions:

- **Attrition Baseline:** Quantifies the breakdown between churned and active customers across the platform.
- **Churn Metrics:** Measures average tenure and total cashback distributed to churned users.
- **Complaint Impact:** Calculates the exact percentage of churned users who registered prior complaints to evaluate service friction.
- **Category & Demographics:** Identifies high-risk product categories (e.g., Laptop & Accessory, Mobile Phone) across city tiers, marital statuses, and genders.
- **Payment Mode Dynamics:** Evaluates payment preferences among active users versus high-frequency ordering groups.
- **Proximity Analysis:** Categorizes customer warehouse distance (Very Close Distance ≤ 5 km up to Far Distance > 15 km) and maps churn distribution across each tier.
- **Post-Purchase Analysis:** Integrates customer_returns transactional logs with churn and complaint records to analyze refund behaviors among dissatisfied users.